

Manifold-Informed Architecture Benchmark

CIFAR100: 100 classes, 3,072D input

Generated: 2026-08-04 03:29:45 | Apple M5 Max MacBook Pro, 64 GB RAM, 2TB SSD

waverider @ 3c5d283 (-abbrev-re 3c5d28344fd5c5ace0b3399593d61b40363afcda) | 2026-08-04 03:22:44
+0000

Best: Intrinsic Dim (PCA→100D→output) test acc = 0.2560 ± 0.0012
vs Standard: +0.0429 (4.29 pp) 184.3× fewer params 221.4× acc/Kparam
Manifold: 3,072D → 19D (99.4% noise dimensions)

Experimental Setup

Parameter	Value
Dataset	CIFAR100
Input dimensionality	3,072D
Classes	100
Intrinsic dim (d)	19
Variance threshold (τ)	0.9
Epochs	100
Trials	3
Batch size	256
Learning rate	0.001
k neighbors (local PCA)	25
Discovery samples	500
Samples per class	10

Manifold Discovery

Local PCA over the training set, $k = 25$ neighbors. Intrinsic dimensionality d is taken as the maximum per-class maximum at $\tau = 0.9$, clamped to $n_{\text{classes}} = 100$.

τ	Mean d	Std	Min	Max	Noise (%)
0.95	18.9	1.0	15	21	99.4
0.90	15.7	1.1	11	18	99.5
0.85	13.3	1.1	9	15	99.6
0.80	11.4	1.1	7	14	99.6

Per-Class Intrinsic Dimensionality

Showing the 10 highest- and 10 lowest-dimensional classes of 100 total, sorted by mean d .

Class	Mean d	Std	Min	Max
motorcycle	18.1	0.5	17	19
bus	17.7	0.5	17	18
butterfly	17.6	0.7	16	18
house	17.6	0.5	17	18
poppy	17.6	0.5	17	18
pickup_truck	17.5	0.5	17	18
tiger	17.5	0.7	16	18
mushroom	17.2	1.0	16	19
orchid	17.1	0.9	15	18
raccoon	17.1	0.8	16	18
...				
plate	14.1	0.8	12	15
apple	14.0	1.3	12	16
bottle	14.0	0.4	13	15
cup	13.7	0.8	13	15
ray	13.5	0.8	12	15
shark	13.0	0.4	12	14
rocket	12.8	0.6	12	14
cloud	12.4	1.1	10	14
plain	12.1	0.9	11	14
sea	11.3	0.8	10	12

Architecture Comparison

Architecture	Params	Test Acc ($\mu \pm \sigma$)	Loss	Acc/Kparam
Standard (1024→512)	3,722,852	0.2131 ± 0.0028	3.7730	0.0001
Wide Manifold (d+1, d=100)	320,573	0.2114 ± 0.0033	3.6483	0.0007
Manifold (d=100)	317,400	0.2089 ± 0.0019	3.5762	0.0007
Manifold + ManifoldAdam (d=100)	317,400	0.2402 ± 0.0064	3.2974	0.0008
ManifoldAdam (1024→512, proj→100D)	3,722,852	0.2193 ± 0.0010	3.5856	0.0001
PCA→100D + MLP (2d→d)	50,400	0.2386 ± 0.0035	3.3877	0.0047
Intrinsic Dim (PCA→100D→output)★	20,200	0.2560 ± 0.0012	3.2191	0.0127

★ = best performing architecture

Key Findings

- **Best architecture:** Intrinsic Dim (PCA→100D→output) — test accuracy 0.2560 ± 0.0012

- **vs Standard:** +0.0429 (4.29 pp) accuracy gain
- **Parameter reduction:** $184.3\times$ fewer parameters (20,200 vs 3,722,852)
- **Parameter efficiency:** $221.4\times$ improvement (0.0127 vs 0.0001 acc/Kparam)
- **Manifold compression:** 3,072D $\rightarrow$ 19D (99.4% of ambient dimensions carry no structure)

Result Figure

Provenance

Item	Value
Repository	Flux-Frontiers/waverider
Commit	3c5d283 (-abbrev-re 3c5d28344fd5c5ace0b3399593d61b40363afcd)
Commit date	2026-08-04 03:22:44 +0000
Commit message	docs(readme): add holographic-rendering Breaking News section; audit fixes
Python	3.12.3
TensorFlow	2.21.0
Device	CPU
Host	vm
OS	Linux-6.18.5-fc-v18-x86_64-with-glibc2.39
Machine	Apple M5 Max MacBook Pro, 64 GB RAM, 2TB SSD
Generated	2026-08-04 03:29:45

CIFAR-100: Manifold-Informed Architecture ($d=100$) vs Standard
3,072D color images $\rightarrow$ manifold discovery $\rightarrow$ architecture | 100 fine-grained classes | run time: 381s | 2026-05-24 13:54

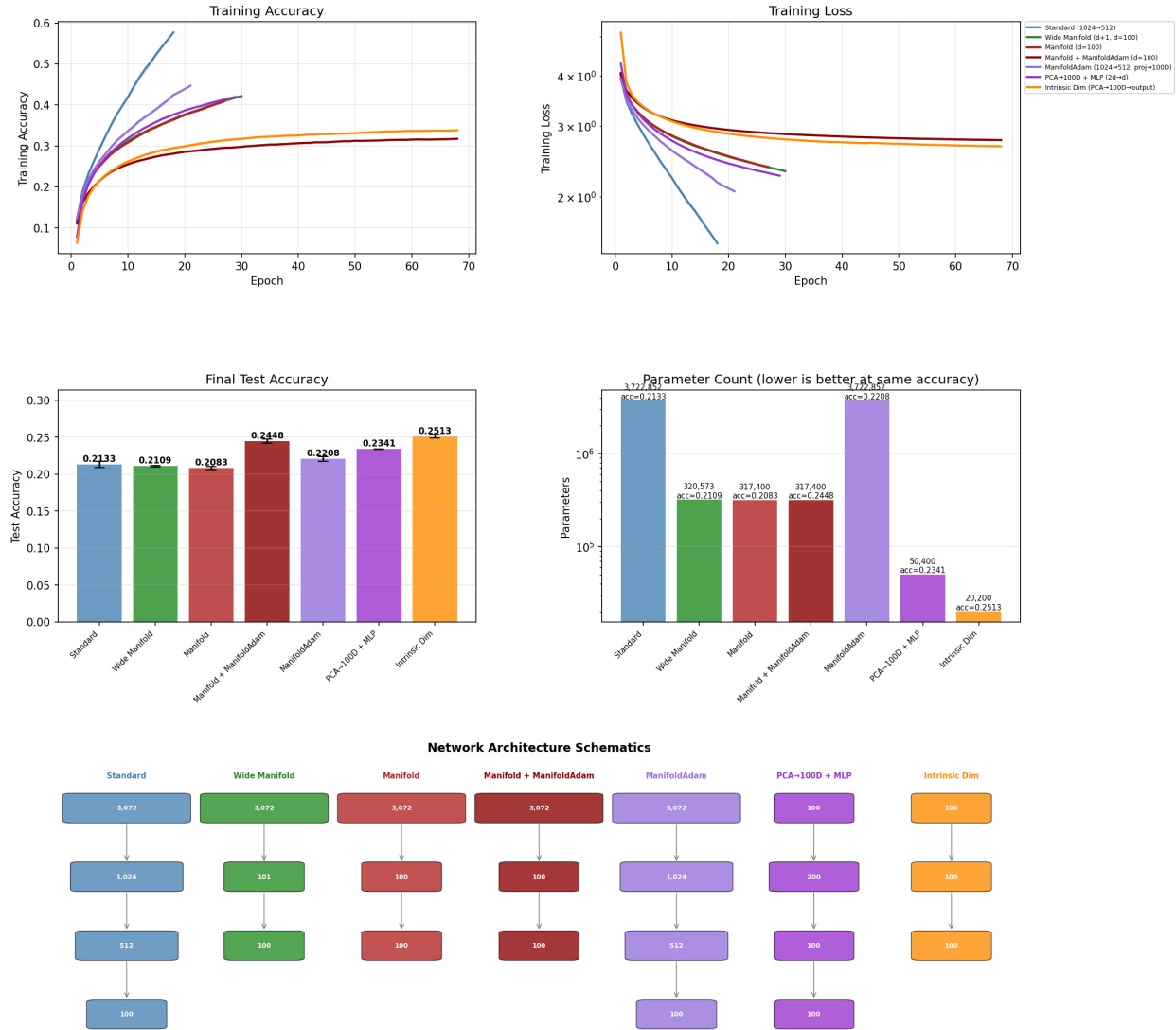


Figure 1: CIFAR100 manifold-informed vs. standard architecture comparison. $d = 19$, 100 epochs, 3 trials.